# Introduction to Deep Learning

From Artificial Intelligence to Deep Neural Networks

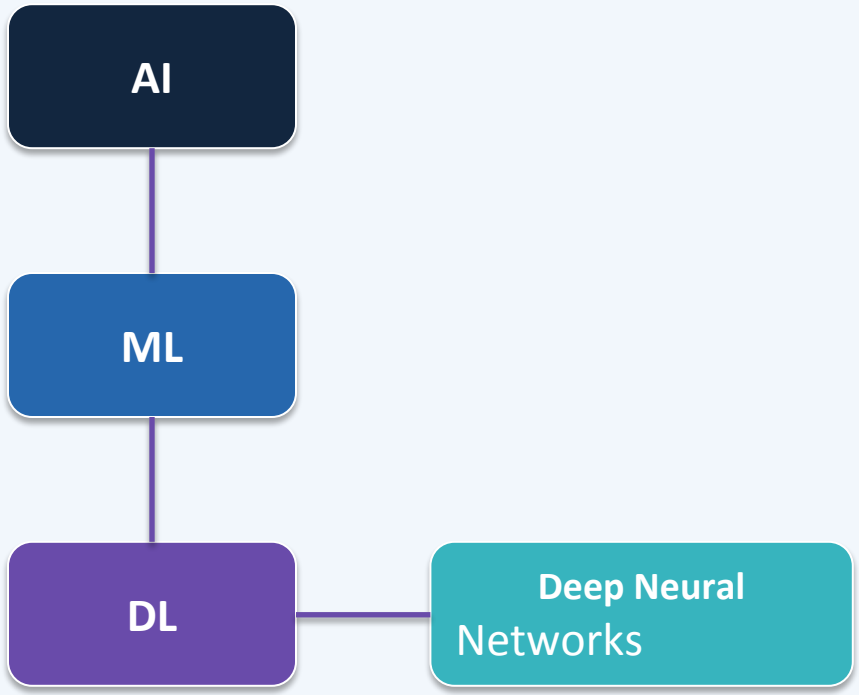
Deep Learning is a subfield of Artificial Intelligence and Machine Learning.

It uses deep neural networks with multiple layers.

Layers learn increasingly complex representations from data.

Inspired by interconnected neurons in the human brain.

Major areas: vision, speech, language, healthcare and autonomous systems.



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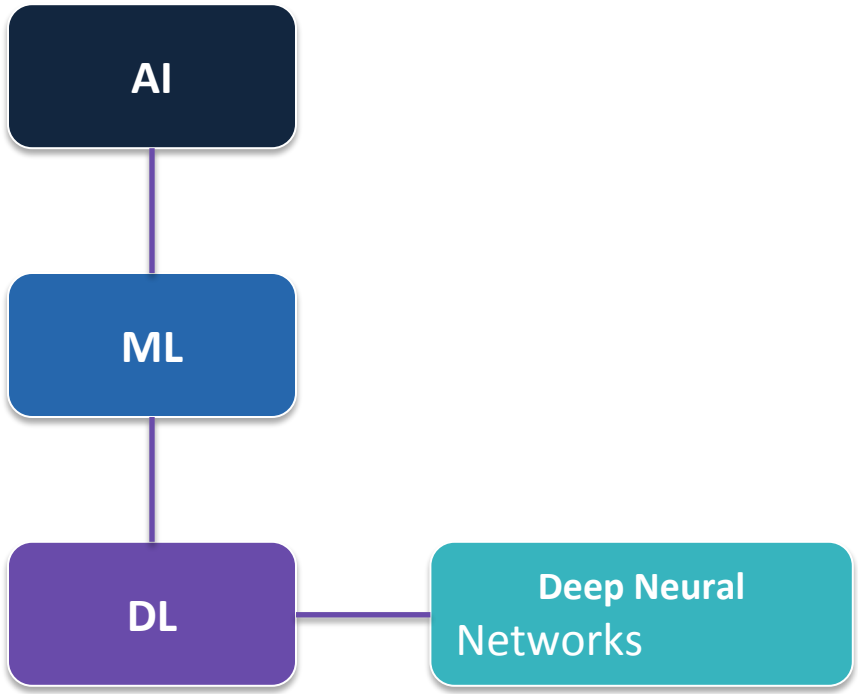
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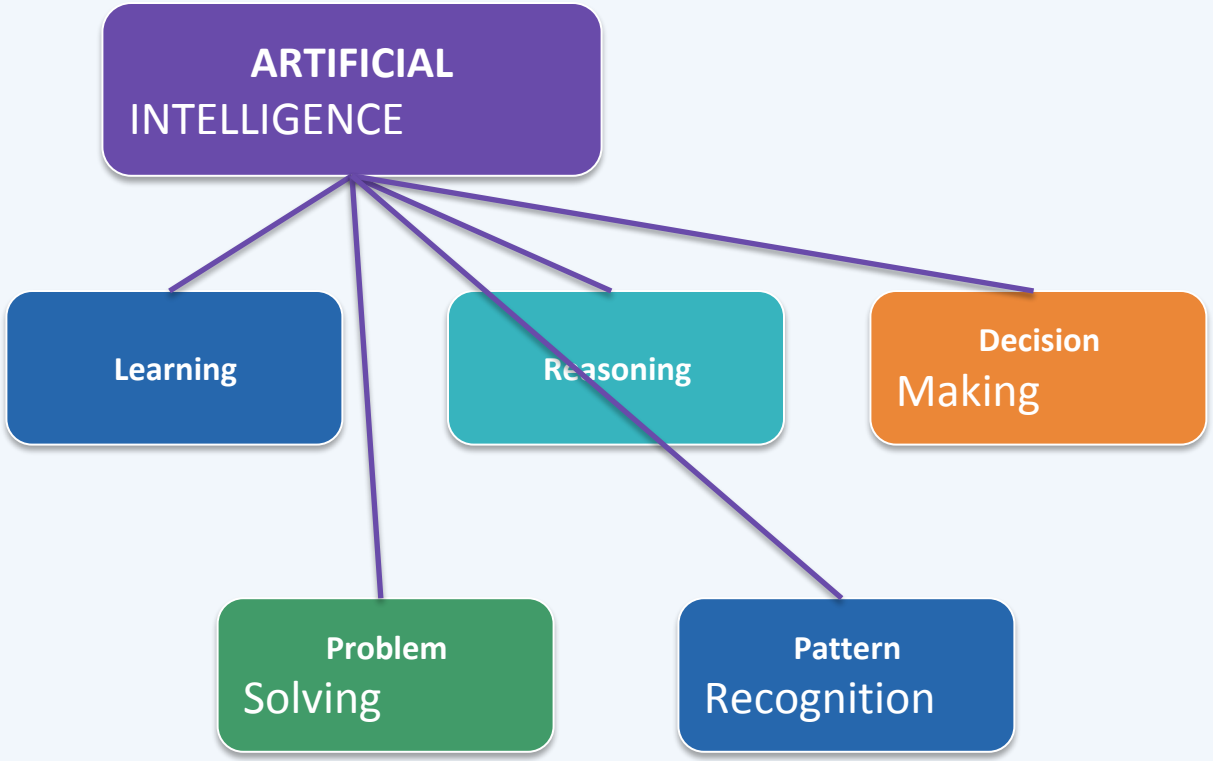
# What is Artificial Intelligence?

The broadest concept

AI gives machines the ability to perform tasks that normally require human intelligence.

Examples include learning, reasoning, decision making, problem solving and pattern recognition.

AI provides the broader foundation for ML and DL.



# What is Machine Learning?

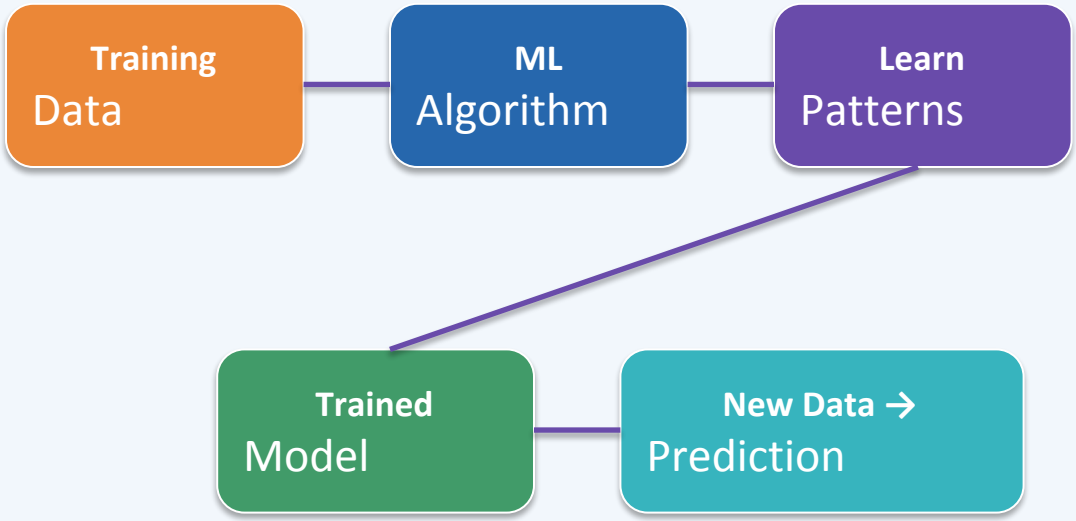
Learning patterns from examples

Machine Learning enables computers to learn patterns from data.

Instead of explicitly programming every rule, the system learns from examples.

Past experience is used to improve performance.

Common tasks include classification, prediction, recommendation and pattern recognition.



# From Machine Learning to Deep Learning

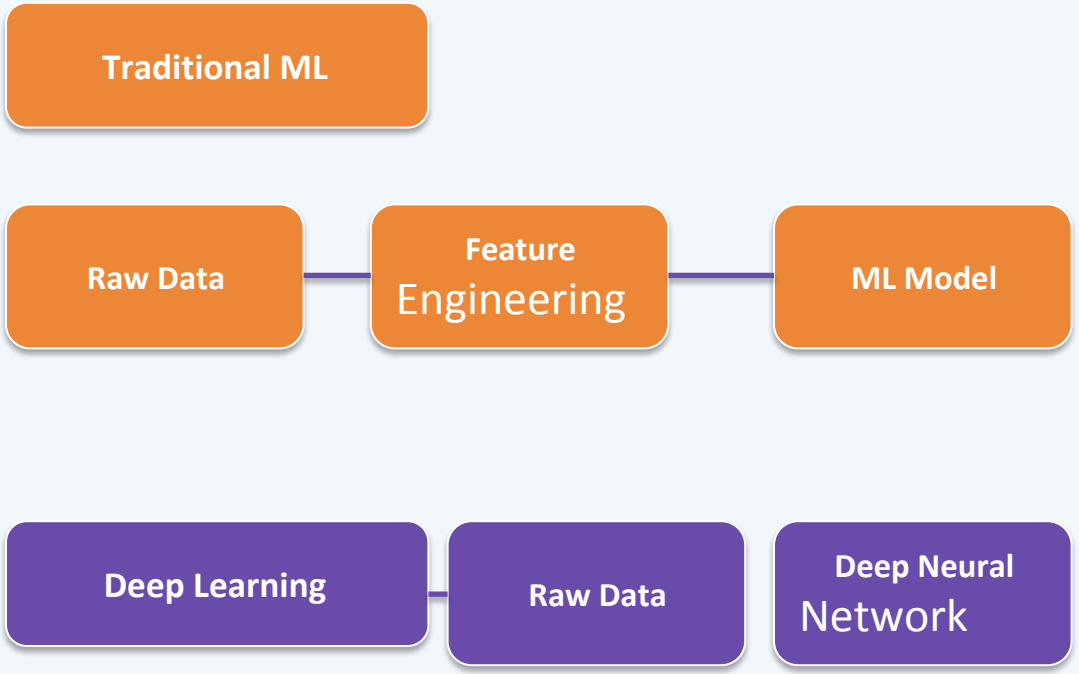
Traditional ML often requires feature engineering.

Deep Learning learns useful features automatically.

Traditional ML can work well with smaller datasets.

Deep Learning generally benefits from large datasets.

Deep models use multiple layers to learn complex representations.



# What is Deep Learning?

Hierarchical feature learning

Deep Learning uses multiple layers of artificial neurons.

Each layer transforms information received from the previous layer.

Early layers learn simple features.

Deeper layers learn increasingly complex features.

This enables complex tasks to be learned from data.




Example: image → edges → shapes → objects → class

# Why is Deep Learning Important?

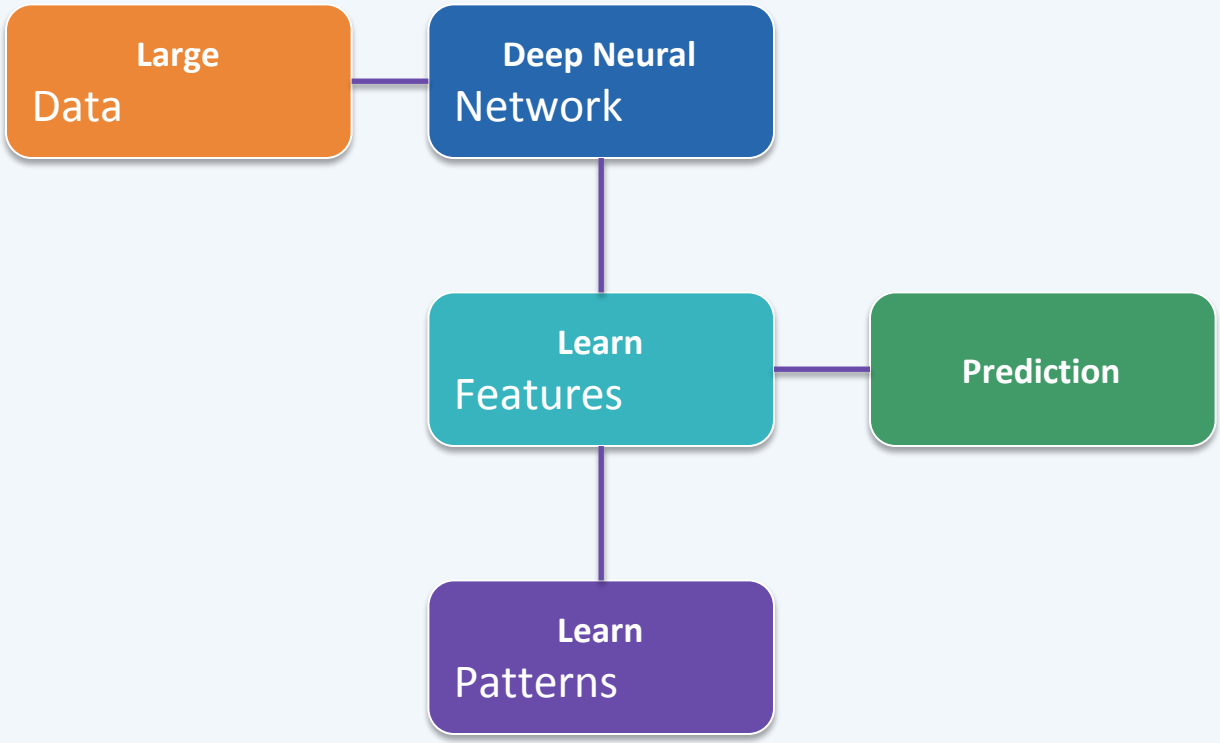
Automatically learns useful features.

Handles complex and high-dimensional data.

Learns hierarchical representations.

Can process images, audio, video and text.

Can improve with sufficient training data.



# Basic Architecture of Deep Learning

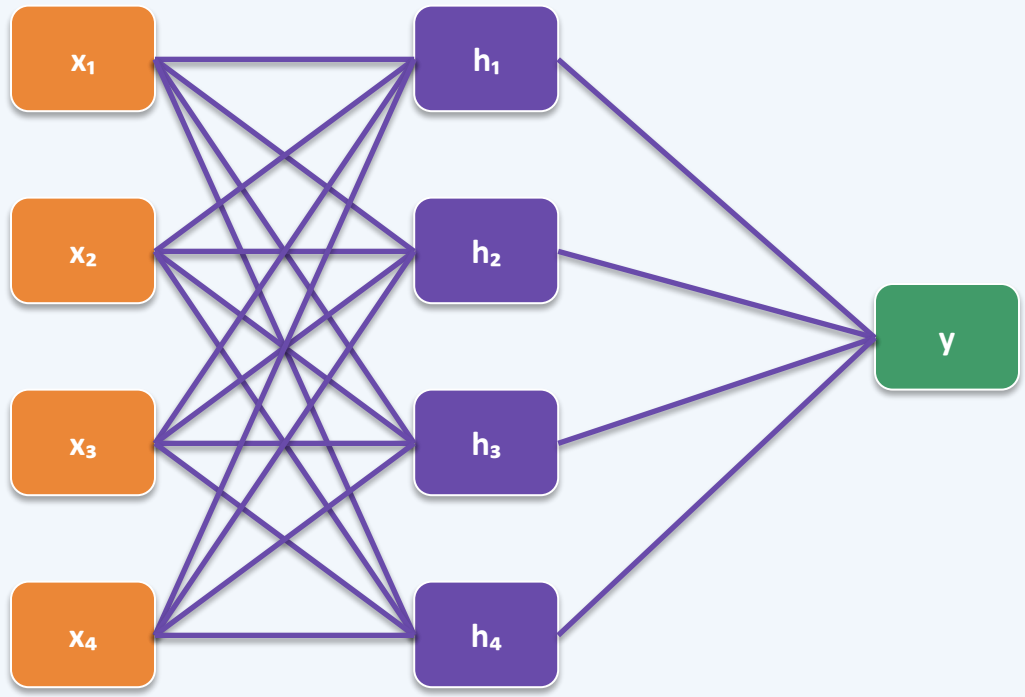
Input → Hidden Layers → Output

Input Layer: receives raw features/data.

Hidden Layer(s): transform data and learn representations.

Output Layer: produces the final prediction.

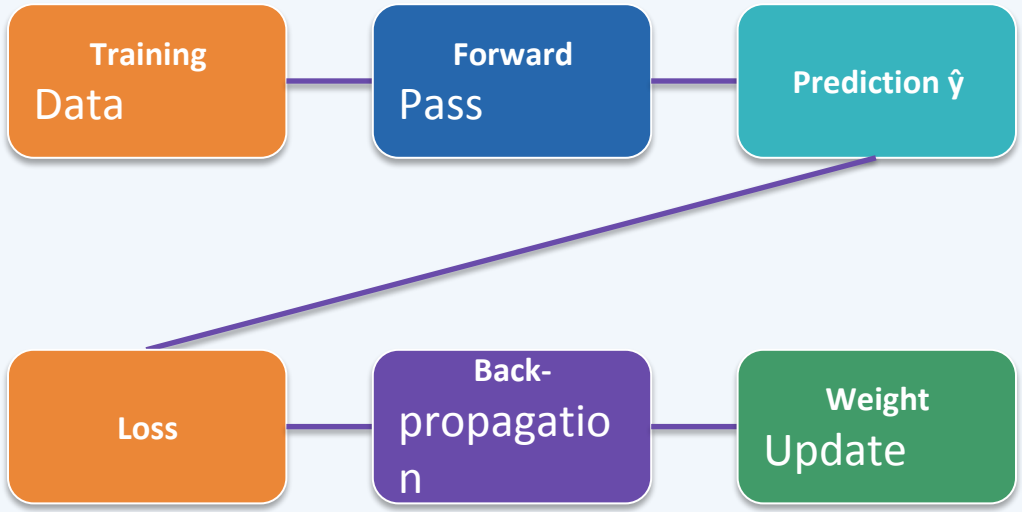
Connections contain weights adjusted during training.



# How Does Deep Learning Learn?

The basic training cycle

- Forward pass produces a prediction.
- Loss measures the difference between target and prediction.
- Backpropagation calculates gradients.
- Weights and biases are updated.
- The process repeats for many epochs.



# Applications of Deep Learning

**Healthcare**

**Cancer detection** from scans

**Automotive**

**Self-driving** cars

**NLP**

**Voice** assistants

**Finance**

**Fraud** detection

**E-commerce**

**Product** recommendation

**Education**

**AI tutors &** personalized learning

# Deep Learning: Summary

AI is the broad field of intelligent machines.

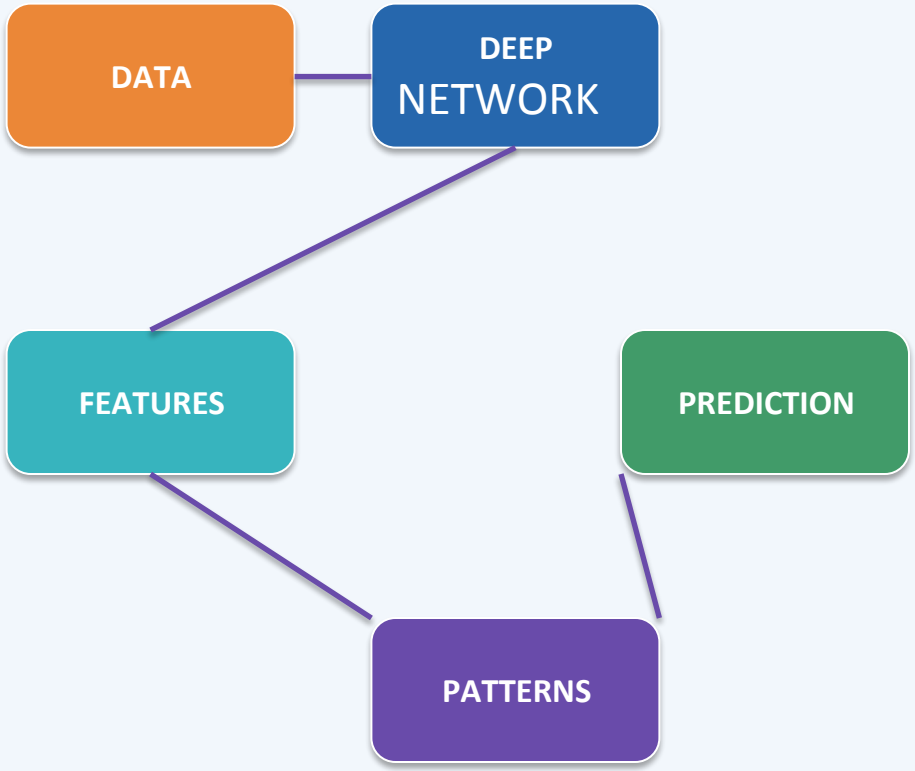
ML enables machines to learn patterns from data.

Deep Learning uses multiple neural-network layers.

Deep networks learn hierarchical features automatically.

Training: Forward Pass → Loss → Backpropagation → Weight Update.

Deep Learning is powerful for images, speech, text and complex data.



# Introduction to Neural Networks

From biological inspiration to computational models

A neural network is a machine-learning model inspired by information processing in the brain.

It consists of interconnected computational units called neurons.

Neurons are organized into layers.

The network learns patterns by adjusting connection weights.



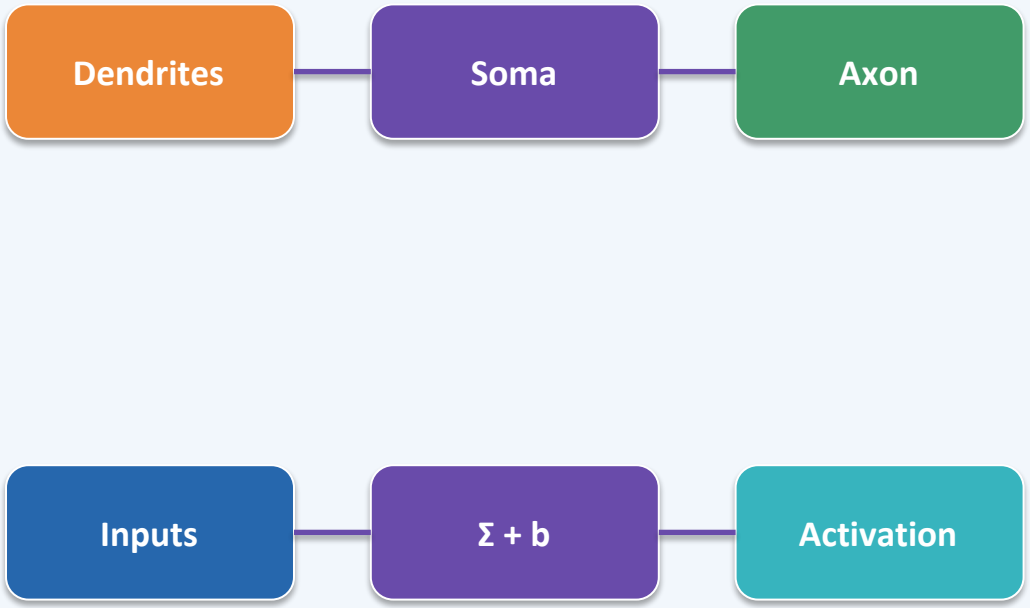
# Biological Neuron → Artificial Neuron

Biological neuron: dendrites receive signals.

Soma/cell body integrates incoming signals.

Axon transmits the resulting signal.

Artificial neurons simplify this idea mathematically using inputs, weights, bias and activation.



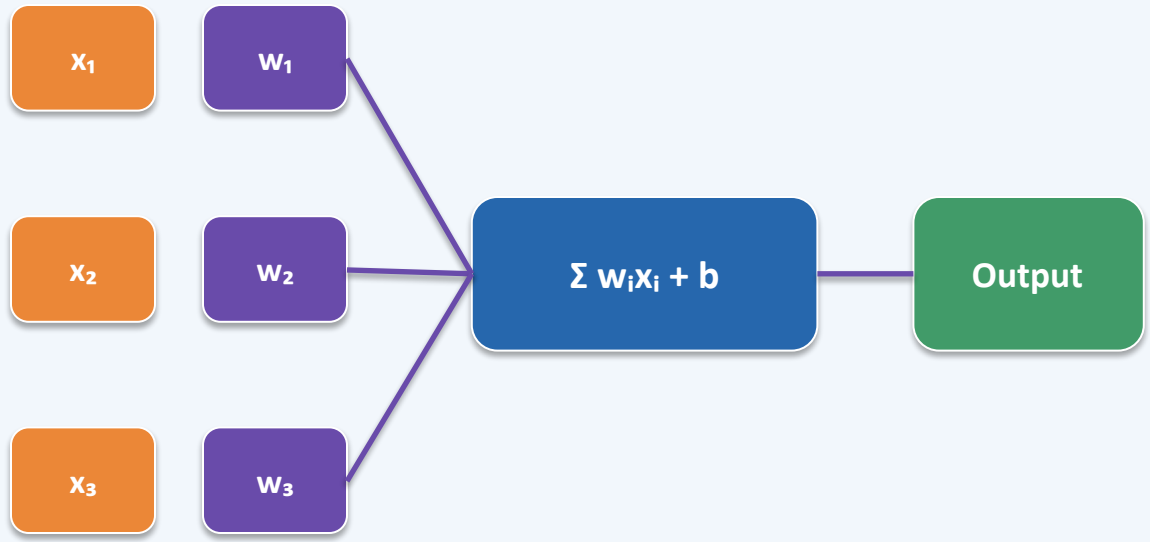
# Basic Building Blocks: Inputs, Weights and Bias

Inputs $x_1$, $x_2$, …, $x$ represent numerical features.

Weights determine the contribution of each input.

Bias shifts the activation threshold or decision boundary.

The neuron first calculates a weighted sum.



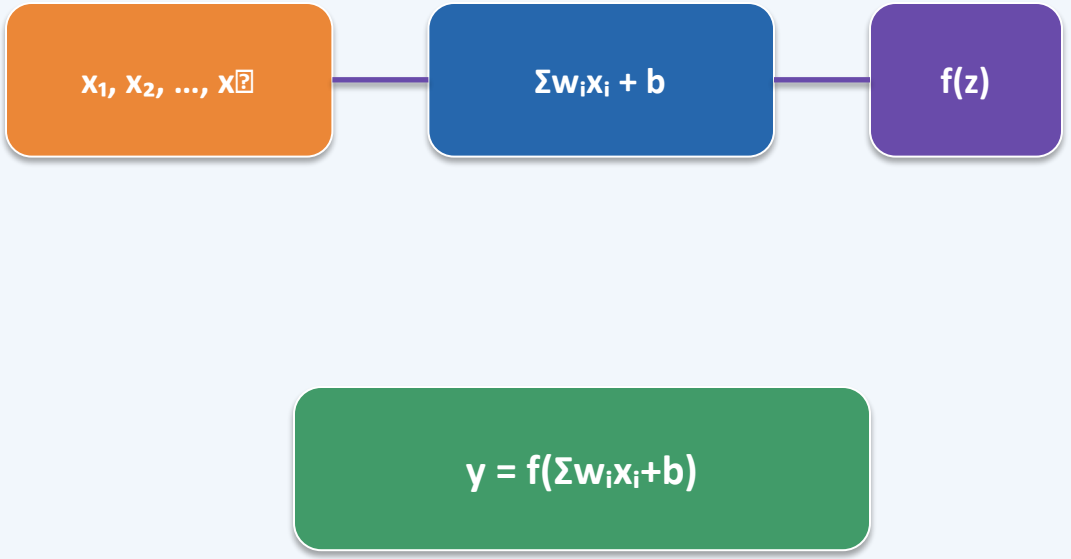
# Artificial Neuron and Activation

The artificial neuron computes a weighted sum plus bias.

The activation function transforms the result.

Activation introduces non-linearity in multilayer networks.

Non-linearity enables complex pattern learning.



# Activation Functions

Step, Sign, Sigmoid, Tanh and ReLU

Step/Sign: basic binary decisions.

Sigmoid: smooth output between 0 and 1.

Tanh: output between −1 and +1.

ReLU: max(0,z), widely used in hidden layers.

Activation functions determine how neurons transform their input.



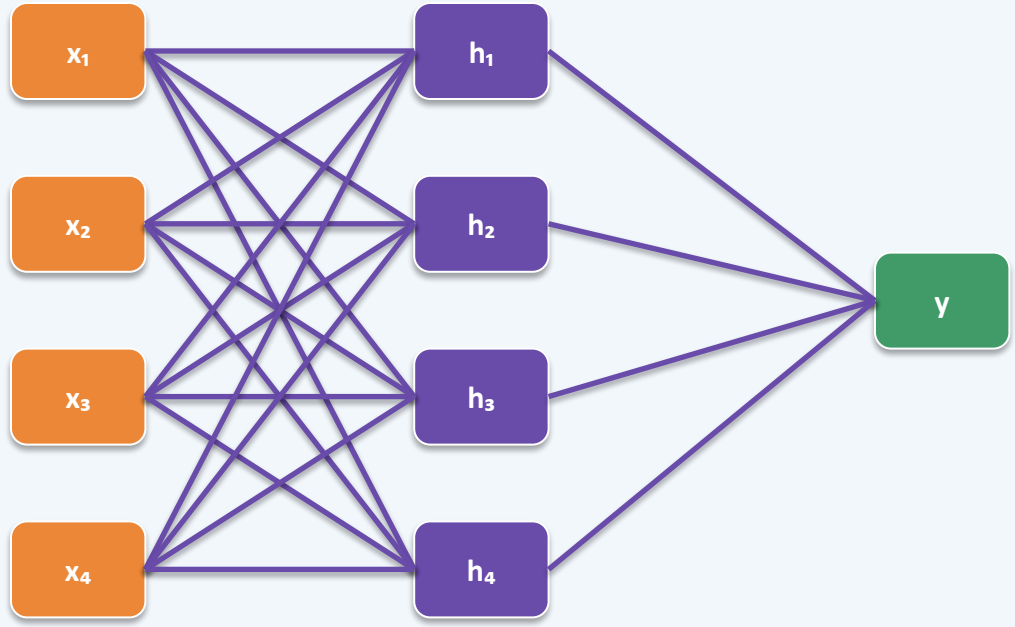
# Neural Network Layers

Input Layer: receives features.

Hidden Layer(s): learn intermediate representations.

Output Layer: produces the final prediction.

More hidden layers create deeper architectures.



# Single-Layer Perceptron

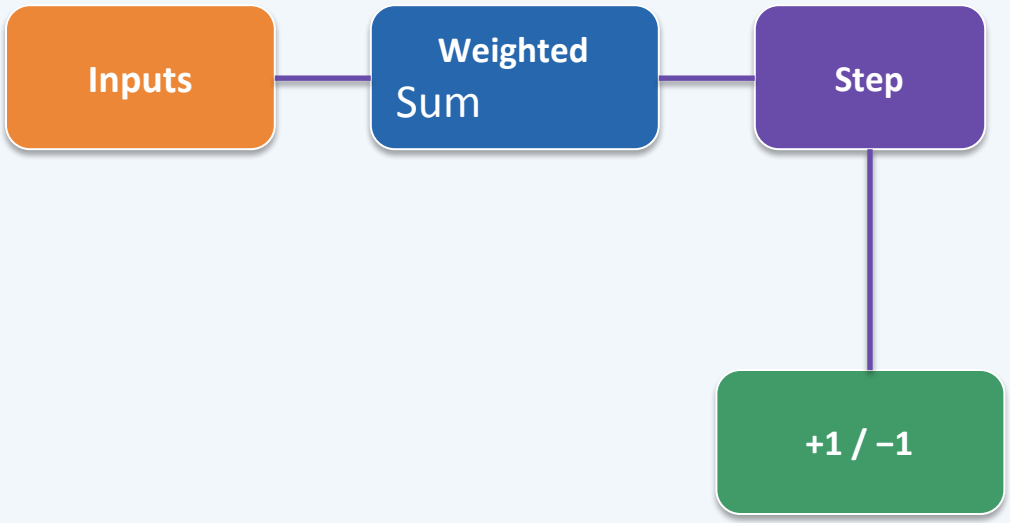
The perceptron is a basic supervised binary classifier.

It uses weighted inputs and a threshold/activation function.

It creates a linear decision boundary.

It can solve linearly separable problems such as AND and OR.

It cannot learn XOR using a single layer.



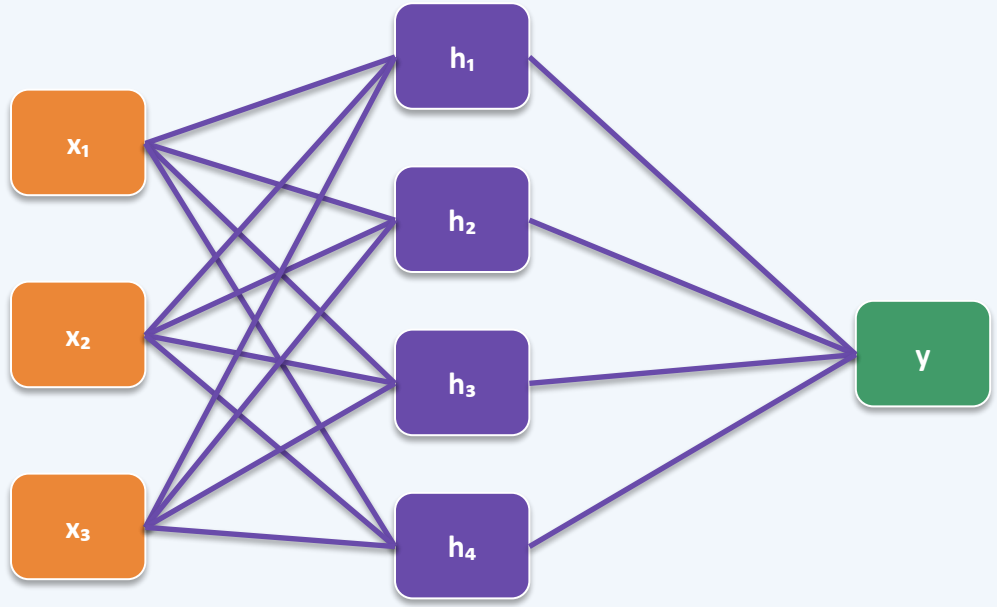
# Multi-Layer Perceptron (MLP)

An MLP contains an input layer, one or more hidden layers and an output layer.

Hidden layers allow the network to learn complex patterns.

Nonlinear activations enable nonlinear relationships.

MLPs are commonly trained using backpropagation.



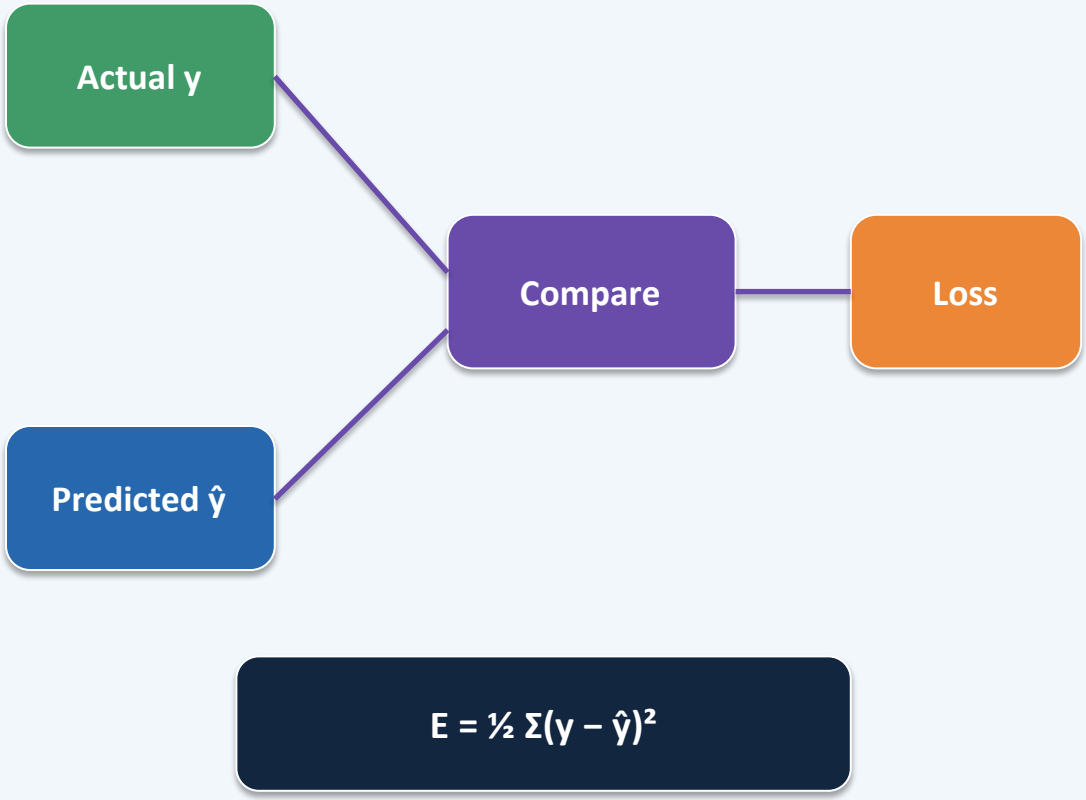
# Loss / Error Function

Prediction is compared with the actual target.

The loss function measures prediction error.

Training attempts to minimize the loss.

One common educational example is squared error.



# Gradient Descent and Learning Rate

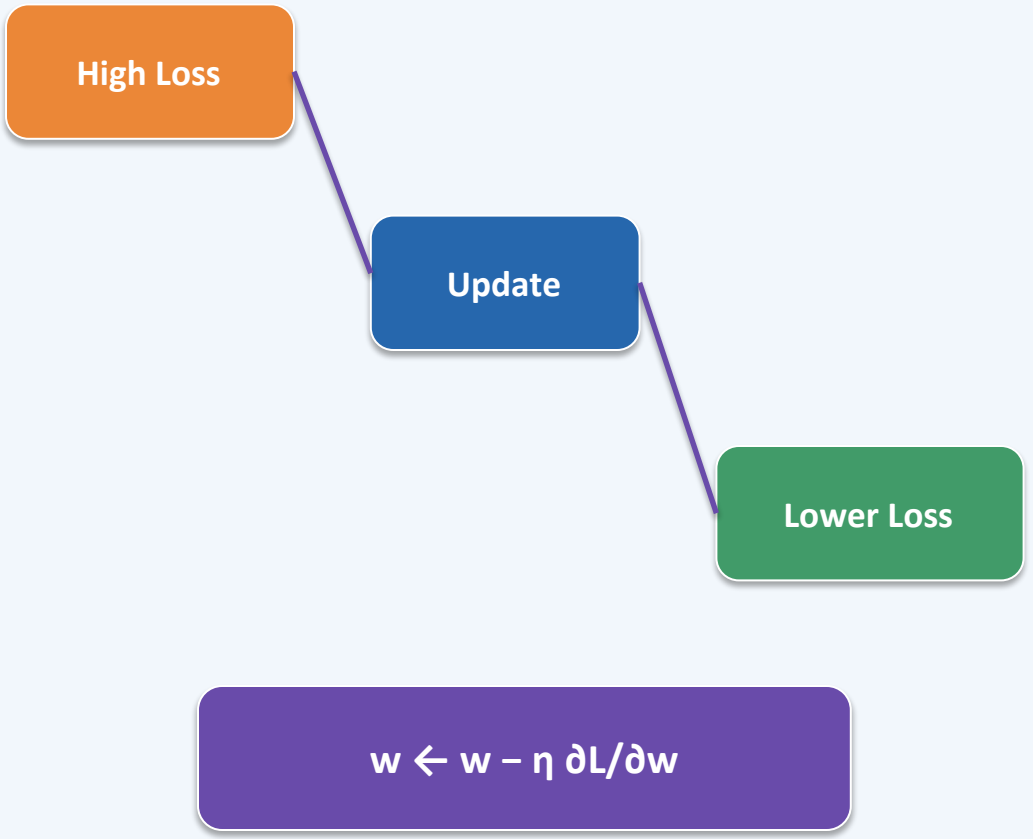
Gradient descent minimizes the loss by adjusting model parameters.

The gradient indicates how the loss changes with a parameter.

The learning rate controls the size of each update.

Too large: unstable or overshooting.

Too small: learning may be slow.



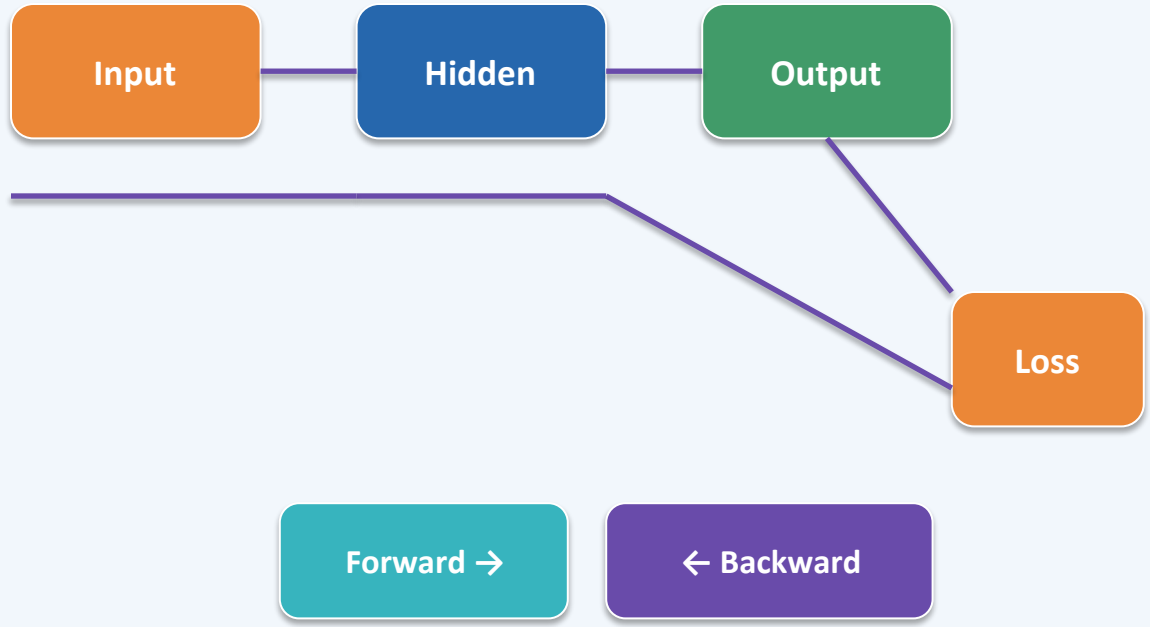
# Backpropagation

Backpropagation computes how the loss changes with respect to weights.

It uses the chain rule of calculus.

Gradients are propagated backward from output toward earlier layers.

Weights and biases are updated to reduce the loss.

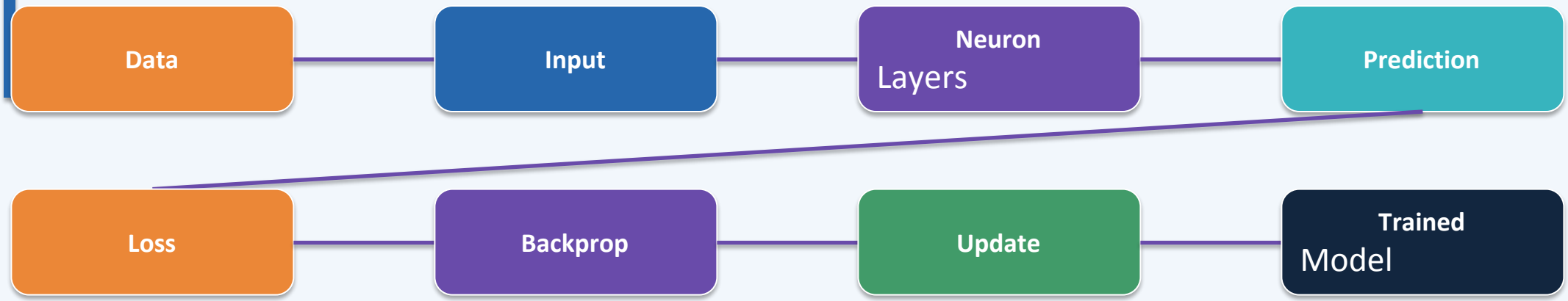


# Complete Neural Network Learning Cycle

The complete cycle combines the building blocks into one training workflow.

Training repeatedly adjusts weights and biases to reduce loss.

The final trained model is then evaluated on unseen data.



# Key points

A neural network is built from interconnected artificial neurons.

Inputs are combined using weights and bias.

Activation functions transform neuron outputs and enable nonlinear learning.

Layers form increasingly powerful representations.

Perceptrons provide the basic foundation; MLPs extend learning with hidden layers.

Training uses loss, gradients, backpropagation and parameter updates.

These concepts form the foundation for modern Deep Learning.

# Training of Neural Networks

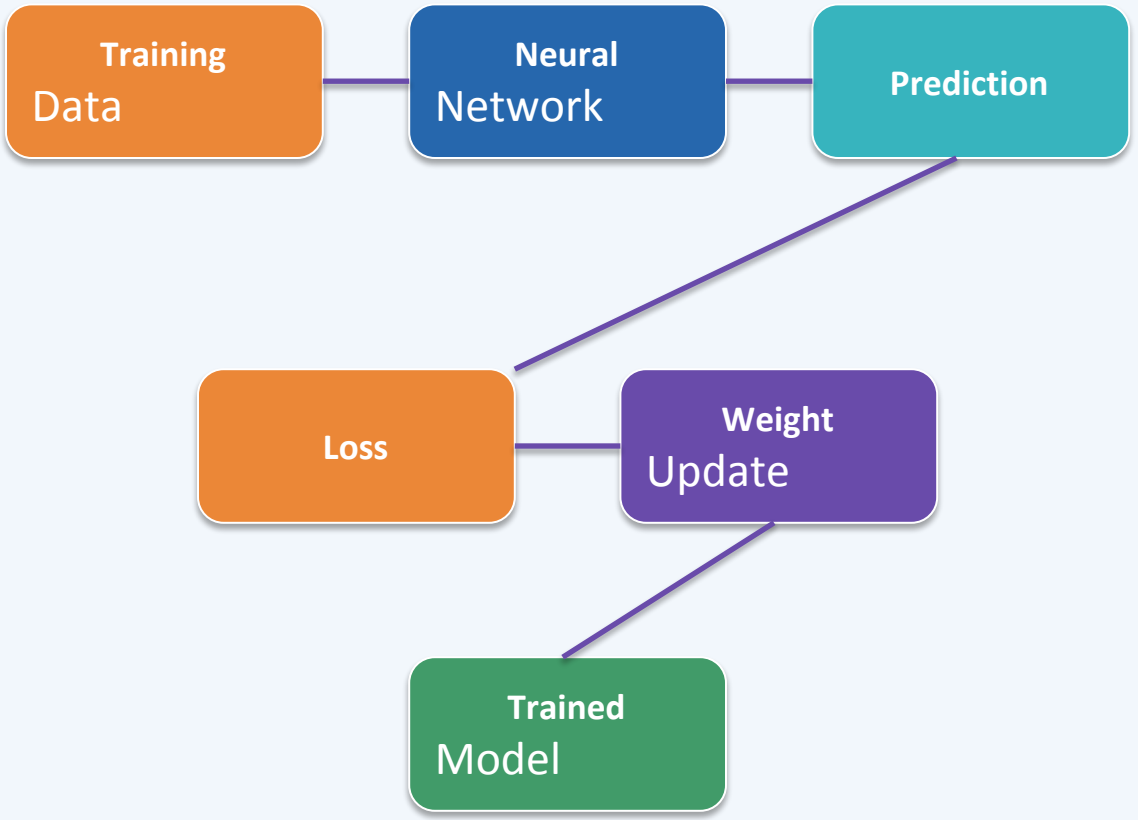
From training data to a trained model

Training is the process of teaching a neural network to learn patterns from examples.

The network adjusts weights and biases based on prediction error.

Training repeatedly performs forward propagation, loss calculation, backpropagation and parameter updates.

The objective is to reduce the loss and improve prediction performance.



# Training Data and Dataset Preparation

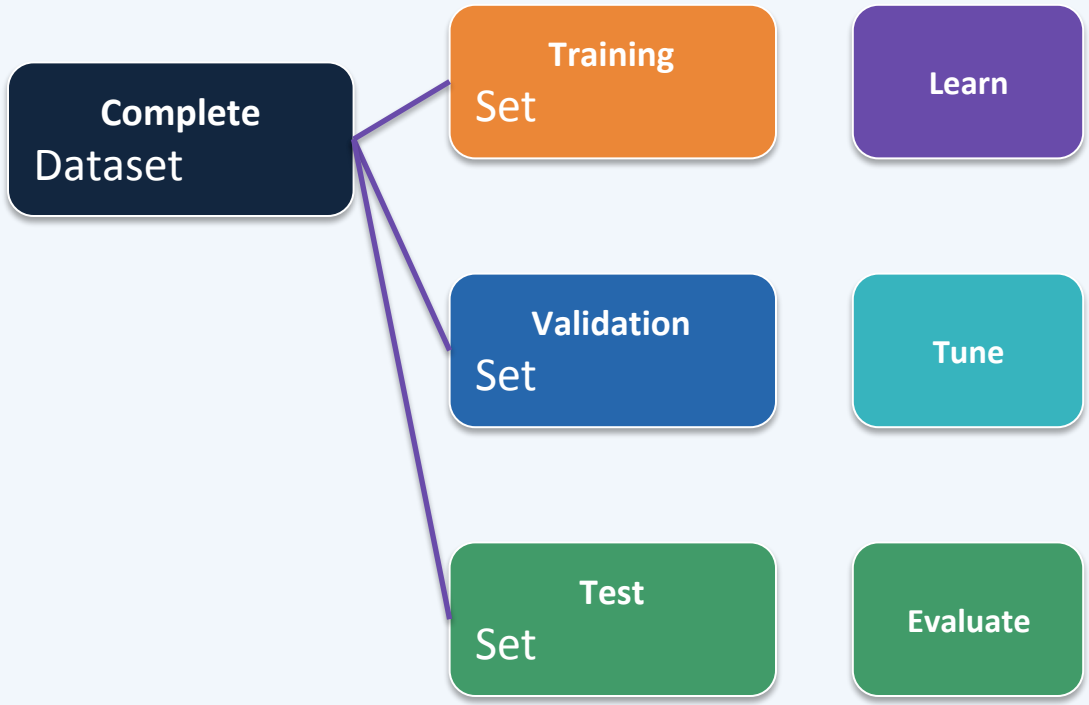
The foundation of neural-network training

Training begins with examples containing input features and, for supervised learning, target outputs.

Data should be organized into suitable numerical representations.

A dataset is commonly divided into training, validation and test sets.

Good-quality and representative data helps the model learn useful patterns.



# Forward Propagation

Generating the network prediction

Input features are passed from the input layer through hidden layers to the output layer.

Each neuron computes a weighted sum plus bias.

An activation function transforms the result.

The final layer produces the prediction ŷ.




$$x \rightarrow z = \Sigma w_i x_i + b \rightarrow f(z) \rightarrow \hat{y}$$

# Loss Function and Error Calculation

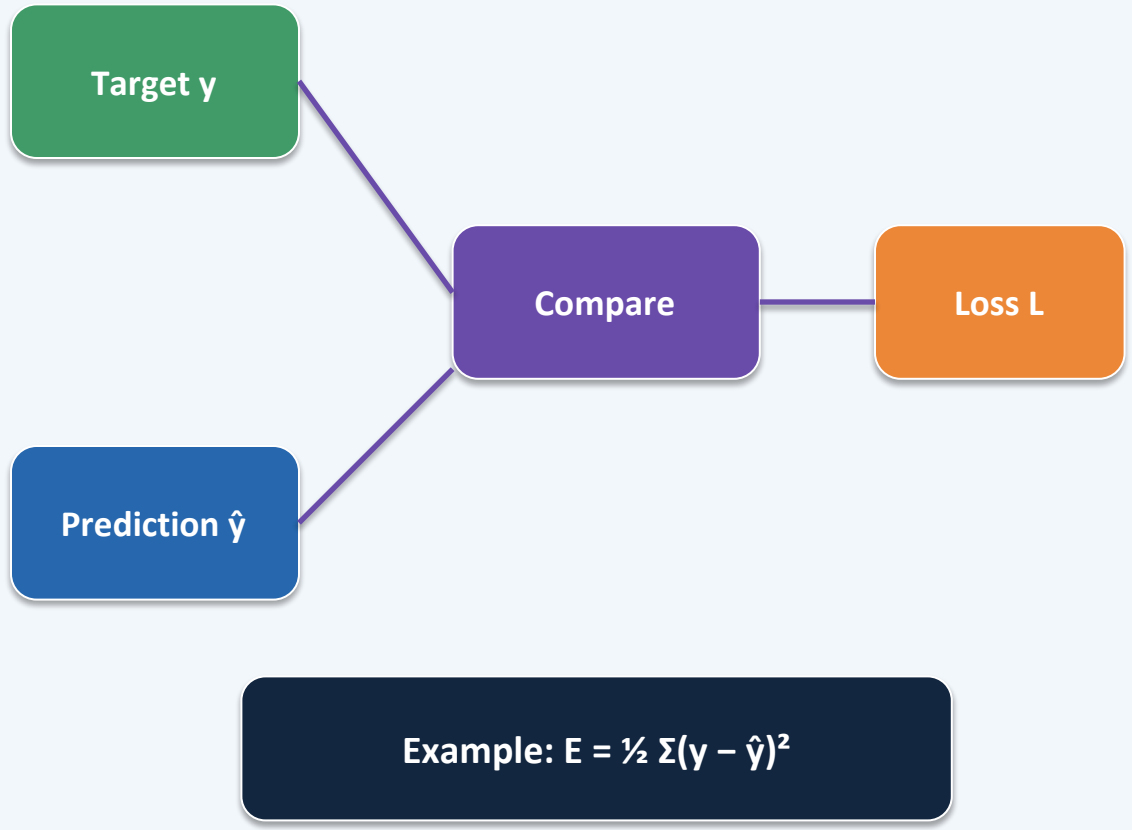
Measuring how well the network performs

The prediction is compared with the target value.

The loss function converts prediction error into a numerical value.

A smaller loss indicates that predictions are closer to the target.

Training aims to minimize the loss.



# Backpropagation

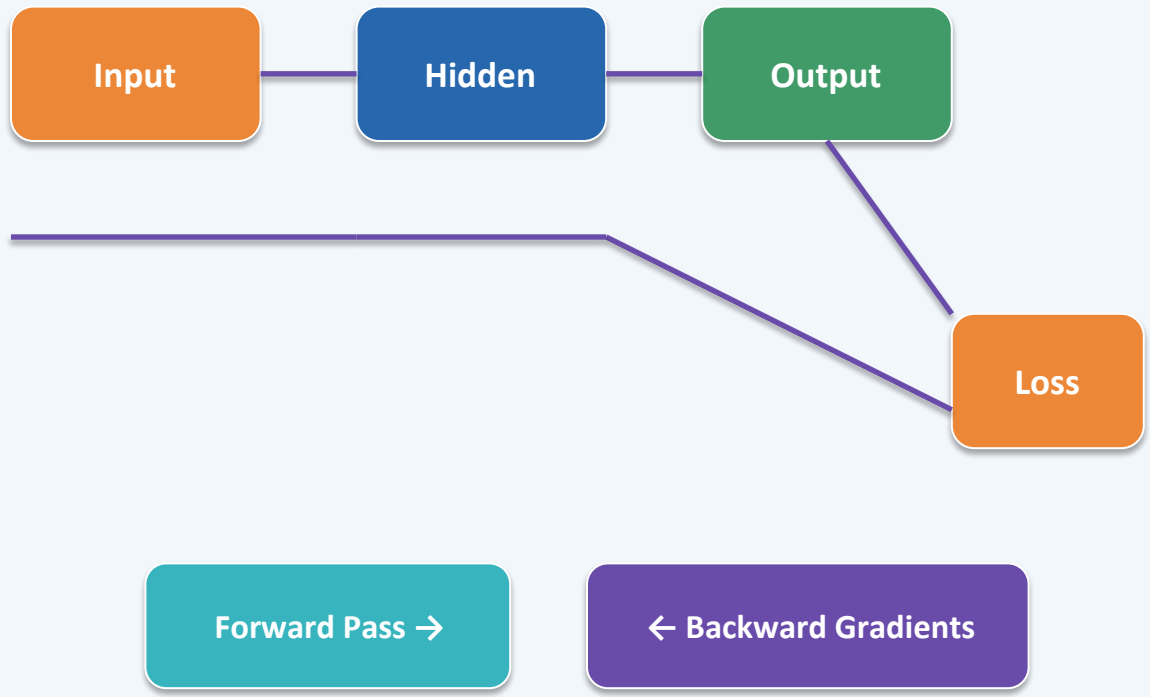
Sending the error information backward

Backpropagation calculates gradients of the loss with respect to network parameters.

It uses the chain rule of calculus.

The error information is propagated from the output layer toward earlier layers.

The resulting gradients guide weight and bias updates.



# Gradient Descent and Weight Updates

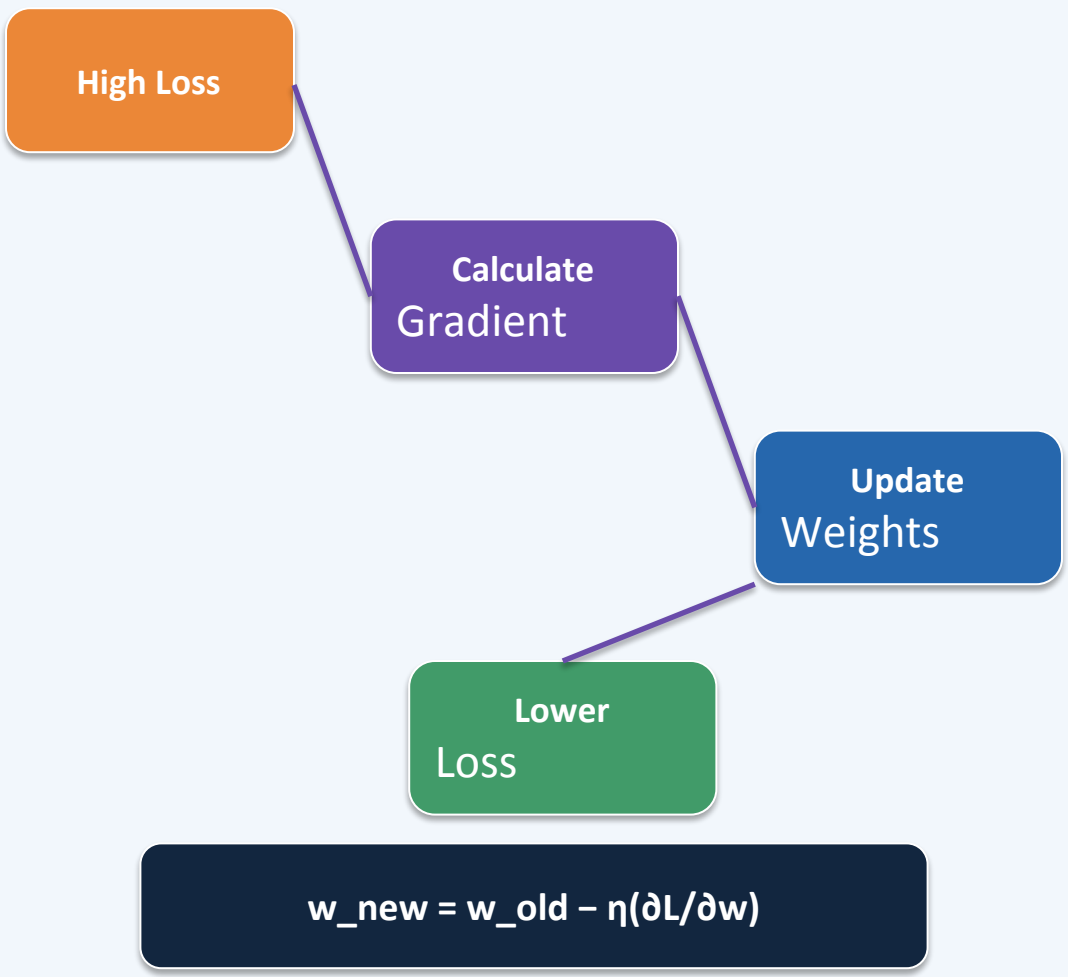
Reducing the loss step by step

Gradient descent is an optimization method used to minimize the loss.

The gradient indicates the direction of increasing loss.

Parameters are updated in the opposite direction of the gradient.

The learning rate η controls the size of each update.



# Epochs, Batches and Learning Rate

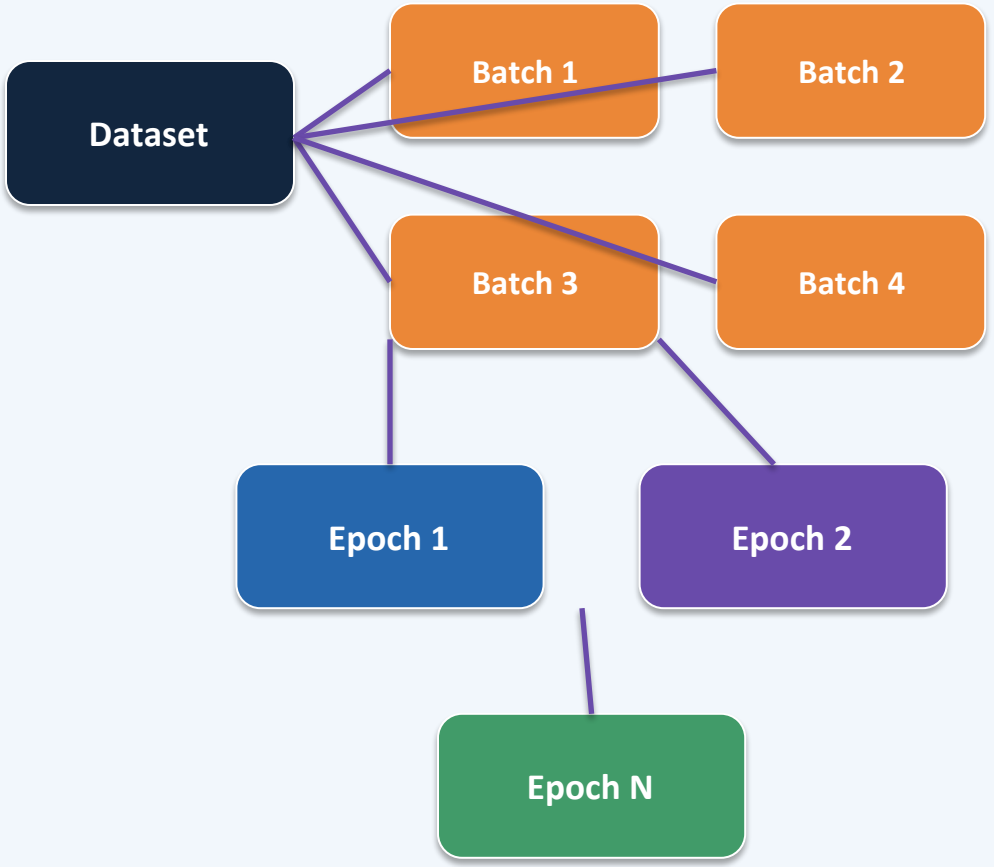
Controlling the training process

Batch: a subset of the training dataset processed in one update.

Epoch: one complete pass through the training dataset.

Learning rate: controls how strongly weights change at each update.

Training normally uses many batches across multiple epochs.



# Complete Training Cycle

From data to a trained neural network

Training combines forward propagation, loss calculation, backpropagation and optimization.

Weights and biases are updated repeatedly over batches and epochs.

Validation data can be used to monitor generalization during training.

After training, the model is evaluated on unseen test data.

